

Machine Learning (CSE-4021) - Course Details

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Machine Learning (course code CSE-4021) is a 3-credit elective course offered to final-year B.Sc. CSE students in Semester 7. The course instructor is Dr. Elena Rodriguez, Professor of Computer Science, and the teaching assistant is Mr. Daniel Kim.

Classes are held on Sunday and Tuesday from 10:00 AM to 11:30 AM in Room 301 of the Technology Building. The weekly machine learning lab session takes place on Tuesday from 2:00 PM to 5:00 PM in the Artificial Intelligence Lab.

The course covers the following major topics: introduction to machine learning and Python tooling, linear and logistic regression, decision trees and random forests, support vector machines, unsupervised learning and clustering, principal component analysis, neural networks, convolutional neural networks for images, recurrent neural networks for sequences, an introduction to transformers, and model evaluation with cross-validation and standard metrics.

The required textbook is Pattern Recognition and Machine Learning by Christopher Bishop. Two recommended references are Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow by Aurelien Geron, and Deep Learning by Goodfellow, Bengio and Courville.

Grading is based on a midterm examination worth 30 percent, a semester-long machine learning project worth 40 percent, and a final examination worth 30 percent. Students must submit the project demo before the 13th week. Attendance below 70 percent disqualifies a student from sitting the final examination.

The prerequisite for this course is Artificial Intelligence (CSE-3201). Students are expected to be comfortable with Python programming and linear algebra. The course uses Python with NumPy, pandas, scikit-learn and PyTorch in the lab sessions.